

Data Flow Diagram

Date	06 July 2026
Team ID	
Project Name	Smart Lender — Applicant Credibility Prediction for Loan Approval
Maximum Marks	2 Marks

DFD Symbol Legend

Symbol	Description
External Entity (Oval)	Applicant, Credit Officer — submit data or receive results
Process (Numbered Rectangle)	An activity that transforms data, e.g. 'Validate Input', 'Predict Loan Status'
Data Store (Solid Rectangle)	Where data is held — loan_prediction.csv, rdf_model.json, scale1.pkl
Labeled Arrow	Movement of data between entities, processes, and stores

Level-0 (Context) Flow

Applicant -> [1.0 Submit Application Form] -> Loan Application Data -> [2.0 Preprocess & Scale Input] -> Scaled Feature Vector -> [3.0 Predict via XGBoost Model] -> Prediction (Approved/Declined) -> [4.0 Render Result Page] -> Credit Officer / Applicant

Data Stores Involved

- D1 — loan_prediction.csv: historical applicant records used to train the model
- D2 — scale1.pkl: the fitted StandardScaler used to normalize new applicant input
- D3 — rdf_model.json: the trained XGBoost model used to generate predictions

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